Gamma-ray Analysis for Multi-Messenger Astrophysics: GAMMA

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CONTENTS

I. Introduction 1

II. Multi-Messenger Astronomy 1

III. Cosmology: GRBs as High-z Probes 1
- A. Cosmological Models and Tensions 1
- B. GRBs as Cosmological Probes 2
- C. Multi-Messenger Cosmology 2
- D. GRB Statistics Estimations 3

IV. Lorentz Invariance and High-Energy Photon Lags 3
- A. A Theory of Quantum Gravity 3
- B. Detecting Lorentz Invariance with High-Energy Photons 3

V. Summary of technical requirements 3

VI. Conclusions 3

Acknowledgments 3

References 4

I. INTRODUCTION

The Gamma-ray Analysis for Multi-Messenger Astrophysics (GAMMA) initiative is a project by the CAPIBARA Collaboration with the goal to study the high-energy, gamma-ray phenomena in the universe and launch a CubeSat to observe these events. In this white paper, we provide a set of compelling open problems that motivate the development and launch of such a mission.

In the 2020 decadal survey [1], it was pointed out that multi-band, that means acros the electromagnetic spectrum, and multi-messenger, that means combining different "messengers" such as light, GW and neutrino, detections were to be prioritized in the next years. Currently operating flagship gamma-ray observatories have long passed their main operation lifetimes, and new replacements are still in very early phases.

Professionally or student developed CubeSats with miniaturized detector technology have proven to be able to successfully detect gamma-ray bursts (GRBs) [2–4]. However, gamma-ray photons cannot be focused as with a traditional optical telescope. A good way around to accurately observing and locating GRBs is to perform triangulation with multiple satellites in orbit. Therefore, having multiple missions with large fields of view and good resolution capabilities is of very much importance for the proper study of this astrophysical phenomena.

Missing clear explanation of what are gamma-ray bursts and why we need to send detectors to orbit?

II. MULTI-MESSENGER ASTRONOMY

III. COSMOLOGY: GRBS AS HIGH-z PROBES

A. Cosmological Models and Tensions

Since the late 1920s, we know that the universe is in expansion [5, 6], this lead to the development of the 'Big Bang theory' later confirmed by the observation of the Cosmic Microwave background (CMB) [7, 8]. In the 1990s, the expansion of the universe was found to be accelerated using Ia SNe observations [9, 10], motivating the description of the expansion rate as a time-dependant parameter $H(t)$ (Hubble parameter), with $H_0 = H(t_0)$. The relation between the distance d_L and the redshift $z \propto v_{\text{radial}}$ is then given by the luminosity distance function:

$$d_L(z, \overline{\Omega}) = (1+z)\frac{c}{H_0}\int_0^{z'} \frac{dz'}{E(z)}$$

where H_0 is the Hubble-Lemaître constant (the present time rate of expansion) and $E(z)$ is the dimensionless Hubble parameter, which is different regarding the cosmological model one considers. The current standard model in cosmology ΛCDM assumes the dominance of cold dark matter (CDM) and a cosmological constant denoted Λ [**?**]. Deriving from the Friedmann equations [**?**] (which is a description of the universe treating its components as fluids) we obtain

$$E^2(z) = \left(\frac{H(t)}{H_0}\right)^2 = \Omega_r a^{-4} + \Omega_m a^{-3} + \Omega_k a^{-2} + \Omega_\Lambda$$

where a is the scale factor set to $a(t_0) = a_0 = 1$ and Ω_i are the energy density parameters for each component of the Universe's energy. According to ΛCDM these are the components contributing to the total energy density of the universe [**?**]:

$$\begin{cases} \text{radiation} : \Omega_r \\ \text{matter} : \Omega_m \\ \text{curvture (geometric)} : \Omega_k \\ \text{dark energy} : \Omega_\Lambda \end{cases}$$

Most of the time, a *flat* ΛCDM model is assummed $\Omega_\Lambda + \Omega_m = 1$ with $\Omega_k = \Omega_r = 0$ (due to flatness and negligible constribution of radiation). When not trying to constraint the cosmology, and assuming a standard cosmological $\Omega_\Lambda = 1 - \Omega_= h_0 = 0.7$ is also assumed, agreeing enough with estimated values for these parameters and $h_0 := H_0/100$.

In recent times, attention has been drawn to the so-called 'Hubble Tension', an open problem in cosmology in which measurements of H_0 don't agree between early-universe CMB and late-time local distance ladder observations [11–13]. More recently, in the era of 'precision cosmology' this tension, alongside the σ_8 [?] and Ω [? ?] tensions have become even more obvious. These tensions alongside the fine-tuning problem of the cosmological constant Λ [?] and the growing evidence of dynamical dark energy (DDE) [?] contrasting previous results and assumptions has motivated the search of alternative methods to measure cosmological parameters and test proposed alternative models.

B. GRBs as Cosmological Probes

Gamma-ray bursts are amongs the most luminous astromical events in the Universe and are visible at large distances, making their potential use as high-z cosmological probes compelling. However, GRBs are not standard candles, and we rely on empirical relations between observables to compute our luminosity distance and thus test different cosmologies. Multiple potential relations to 'standardize' GRBs have been proposed [?].

When utilizing GRBs to probe the cosmological parameters, we encounter the circularity problem, which is that we need to assume a cosmology $\overline{\Omega}$ in order to compute the fit to the empirical relation, which we want to use to constraint the parameters. For example, the Amati relation relates the intrinsic spectral peak energy (E_{peak}) and the isotropic equivalent energy (E_{iso}) and is written as:

$$E_{\text{p,i}} = a \cdot E_{\text{iso}}^b$$

this can be used to contraint the cosmology since E_{iso} is computed from the observed fluence $\mathcal{F}$ relying on the luminosity distance by

$$E_{\text{iso}}(\overline{\Omega}) = 4\pi \mathcal{F} d_L(z, \overline{\Omega})/(1+z)$$

In this case, in order to compute (a, b) we need to assume $\overline{\Omega}$. The circularity problem is avoided by computing the empirical relation fit for a set of possible cosmologies, among which the 'best cosmology' is assumed to be the one with the least scatter in the empirical relation.

Key instrumentation requirements for this purpose is a broad energy range, including high-energy γ-ray sensitivity and enough energy resolution, allowing to accurately measuring E_{peak}. Additionally, a broad field of view contributes to increasing the chances of detecting this short-lived transient events, more data . According to estimates [?], 0000 detections with average uncertainty of $\sigma_E = 0.00$ are required to constrain cosmological parameters to a 10% error with 99% certainty.

While GRBs are sensitive to dark energy and matter energy density parameters fairly well, since there are no close anchors, GRBs do badly constraint H_0, indeed it's introduction as a free parameter brings in a degeneracy with the empirical relation. That is why GRBs are better used as a complementary information to another probe, such as GW luminosity distance data. [?]

how much energy resolution for $X\%$ error on energy? What about other empirical relation's requirements?

Key instrument requirements for this purpose include a broad energy range (typically from 10keV up to several MeV) to ensure the spectral peak energy $E_{\text{p, i}}$ is not missed, alongside a sufficient energy resolution ($\Delta E/E \approx 10 - 20\%$) to accurately fit the spectrum. Additionally, a broad field of view contirbutes to increasing the chances of detecting these short-lived transient events, drastically improving the statistical power of the cataloag. According to Fisher matrix forecasts, roughly 800 to 1,000 detections with an average measurement uncertainty of 0.10 dex (which translates to approximately 25% error on the energy observables) are required to constrain the matter density parameter Ω_m to a 10% precision level using the Amati relation. Reaching a 1% precision scale requires a sample sizes in the tens of thousands due to the hard mathematical floor established by the intrinsic scatter of these relations.

While the Amati relation relies solely on prompt gamma-ray emission, other empirical relations can tighten this intrinsic scatter but pose stricter multi-wavelength requirements. For instance, the Ghirlanda relation connects the peak energy to the collimated true energy (E_γ) and features a significantly lower intrinsic scatter (≈ 0.10 dex). However, utilizing it requires precise X-ray or optical follow-up to observe the afterglow jet break time and calculate the opening angle θ_j. Because GRBs lack low-redshift anchors, they struggle to independently constrain the Hubble constant H_0, creating a degeneracy with empirical relation's zero-point. Therefore, GRBs are most powerful when utilize as complementary probes alongside gravitational wave luminosity distance data.

C. Multi-Messenger Cosmology

what does GW work so well together with GRB when constraining cosmologies? first write II, since it build on that

D. GRB Statistics Estimations

In order to compute cosmological parameteres with accuracy a large sample of GRBs is needed, so that the fit of the empirical relation and over the cosmological parameter space has as lowest error uncertainty as possible. We compute the number of GRBs necessary to observe with average measuremeng uncertainty ($0.01\text{dex} < \sigma_{\text{obs}} < 0.2\text{dex}$) to obtain 10% and 1% precision on Ω_m. For the the error uncertainty propagation we compare 3 different standard empirical relations: Amati, Yonetoky, Ghirlanda relation [**?**].

- Amati: $E_{\text{p,i}} \propto E_{\text{iso}}^a$
- Yonetoku: $E_{\text{p,i}} \propto L_{\text{iso}}^b$
- Ghirlanda: $E_{\text{p,i}} \propto E_\gamma^c$

Where a, b, c are free parameters, L_{iso} is the isotropic Luminosity, which is related to the luminosity distance by $d_L(z) = (L_{\text{iso}}/4\pi F)^{1/2}$, and E_γ is the collimation corrected energy, $E_\gamma = E_{\text{iso}}(1 - \cos\ \theta_j)$ with θ_j being the jet opening angle. E_γ can only be computed if θ_j is known, i.e., if the jet break time t_{break} is observed in the afterglow, thus highlighting the importance of rapid alerts and follow-up observations in multiple energy ranges.

Since GRBs are supposed to be jet-drive phenomena, due to relativistic aberration an observer only 'see' photons within a cone of aperture defined by $\theta_\Gamma \propto \Gamma^{-1}$ where Γ is the bulk Lorentz factor of the emitting material. This emitting material gets slowed down by the medium around the GRBs source, thus as the jet travels, Γ diminishes and θ_Γ increases. During this time the observer perceives increasing luminosity, since the cone diameter gets larger, and then a peak and an afterglow as the emission decreases. The turning point happens when $\theta_\Gamma = \theta_j$, then the totality of the jet's surface is visible and there is a sudden decrese in luminosity. If we approximate the afterglow emission as al inear function over time, then the slope becomes smaller.

To compute the estimations on the sample size for each realtion, we employ the Fisher information matrix for any relation $X - Y$ we write, Y corresponds to a different obserable in each relation (see. Table I.

$$\log_{10} Y = a + b \log_{10} E_{p,i}$$

where Y is mathematically related to $d_L(z)$, we shall write

$$Y = 4\pi d_L^2(z) \cdot X_{\text{obs}} \cdot (1+z)^\alpha$$

Given the distance modulus

$$\mu(z) = 5 \log_{10}\left(\frac{d_L(z)}{\text{Mpc}}\right) + 25$$

Then by substituting into the distance modulues we find

$$\mu(z) = \frac{5}{2}\left[a + b\log_{10} E_{\text{p,i}} - \log_{10} X_{\text{obs}} - \alpha \log_{10}(1+z) - \log_{10}(4\pi)\ \right] + 25.$$

Then, applying standard error propagation taking into account the intrinsic physical scatter σ_{int} together with the measurement uncertainties follows:

$$\sigma_\mu = \frac{5}{2}\sqrt{\sigma_{\text{int}}^2 + b^2\sigma_{\log E_{\text{p,i}}}^2 + \sigma_{\log X_{\text{obs}}}^2}$$

TABLE I. Gamma-Ray Burst Relations

Relation	**Y**	$\mathbf{X}_{\text{obs}}$	α
Amati	E_{iso}	S_{bolo} (bolometric fluence)	-1
Ghirlanda	E_γ	$S_{\text{bolo}}(1 - \cos\theta_j)$	-1
Yonetoku	L_{iso}	P_{bolo} (peak flux)	0

The intrinsic error respresents a fixed, intrinsic variation in the empirical relation. It ranges from $\sigma_{int} \approx 0.10$dex for Ghirlanda to $\sigma_{\text{int}}0.30$dex for Amati and Yonetoku. In order to project these errors into the computation of a cosmological parameters such as Ω_m, we calculate the Fisher Information Matrix F.

IV. LORENTZ INVARIANCE AND HIGH-ENERGY PHOTON LAGS

A. A Theory of Quantum Gravity

- briefly explain why a theory of quantum gravity is of such relevance

B. Detecting Lorentz Invariance with High-Energy Photons

- elaborate on how detecting high-energy photon logs provides proof of a granular (quantuized) space-time
- transform the theory into instrumentation requirements (spectral range and resolution, timing resolution)

V. SUMMARY OF TECHNICAL REQUIREMENTS

Section: COSMOLOGY: GRBS AS HIGH-z PROBES

- high enough energy and spectral resolution to accurately identify E_{peak}
- broad energy range (especially towards higher energies) to don't be blind to E_{peak}
- broad field of view (to observe more GRBs)

VI. CONCLUSIONS

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